Please check the examination details below before entering your candidate information

Candidate surname	Other names

Centre Number

Learner Registration Number

Pearson BTEC Level 3 Nationals Certificate, Extended Certificate, Foundation Diploma, Diploma, Extended Diploma

Tuesday 23 May 2023

Morning (Time: 40 minutes)

Paper reference **31617H/1C**

Applied Science/Forensic and Criminal Investigation

UNIT 1: Principles and Applications of Science I

Chemistry

SECTION B: PERIODICITY AND PROPERTIES OF ELEMENTS

You must have:
A calculator and a ruler.

Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and learner registration number.
- Answer **all** questions.
- Answer the questions in the spaces provided
  – *there may be more space than you need.*

Information

- The exam comprises three papers worth 30 marks each:
  – Section A: Structures and Functions of Cells and Tissues (Biology)
  – Section B: Periodicity and Properties of Elements (Chemistry)
  – Section C: Waves in Communication (Physics).
- The total mark for this exam is 90.
- The marks for **each** question are shown in brackets
  – *use this as a guide as to how much time to spend on each question.*
- The periodic table of elements can be found at the back of this paper.

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over

Pearson

Answer ALL questions. Write your answers in the spaces provided.

Some questions must be answered with a cross in a box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

1 Figure 1 shows the symbols of elements present in period 3 of the periodic table.

Na	Mg	Al	Si	P	S	Cl	Ar

Figure 1

(a) What do the atoms of the elements of period 3 have in common?

(1)

☒ **A** same number of protons in the nucleus

☒ **B** same number of neutrons in the nucleus

☒ **C** same number of occupied electron shells

☒ **D** same number of electrons occupying the outer shell

(b) Give the symbol of **one** of the elements in period 3 that can form covalent bonds with oxygen.

(1)

....

(c) Give the symbols of **two** elements in period 3 that are in the s-block of the periodic table.

(2)

.... and ....

(d) Give the symbol of the most electronegative element in period 3.

(1)

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(e) Explain the trend in melting point from sodium to aluminium.

(4)

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(Total for Question 1 = 9 marks)

2 Ammonia, NH_3, is a simple molecular compound.

(a) Hydrogen bonding occurs between molecules of ammonia.

Figure 2 shows two molecules of ammonia.

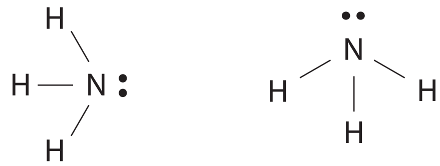

Figure 2

Draw **one** hydrogen bond between the two molecules shown in Figure 2.

You should include the partial charges on the molecules.

(2)

(b) Ammonia can be reacted with acids to form salts.

The salts formed can be used as fertilisers.

(i) Ammonia reacts with sulfuric acid to form the salt ammonium sulfate.

Identify the correct formula for ammonium sulfate.

(1)

☒ **A** NH_3SO_4

☒ **B** NH_4SO_4

☒ **C** $(NH_4)_2SO_4$

☒ **D** $NH_4(SO_4)_2$

(ii) The ammonium ion contains a dative covalent bond.

Describe what is meant by a **dative covalent bond**.

(2)

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....

(c) (i) Ammonium nitrate is another salt of ammonia that can be used as a fertiliser.

It can be made from the reaction of ammonia with nitric acid.

The balanced equation for the reaction is

$$NH_3 + HNO_3 \rightarrow NH_4NO_3$$

A manufacturer reacts 500 g of ammonia with 1000 g of nitric acid.

Determine the limiting reagent in the reaction.

(relative formula masses NH_3 = 17.0, HNO_3 = 63.0)

Show your working

(2)

limiting reagent = ...

(ii) A student wants to carry out the same reaction in the laboratory on a smaller scale.

The student uses 50 cm^3 of a 0.3 mol dm^{-3} solution of ammonia.

Calculate the mass of ammonia in the solution in grams.

(relative formula mass NH_3 = 17.0)

Show your working

(3)

mass of ammonia = ... g

(Total for Question 2 = 10 marks)

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3 Figure 3 shows a table of the elements in period 2 and the relative size of their atoms and ions.

The table is not complete.

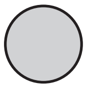
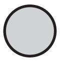
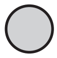
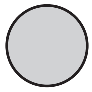
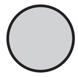
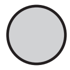
atom	Li	Be	B	C	N	O	F	Ne
atomic radius								
ion	Li^{+}	Be^{2+}	B^{3+}	C	N^{3-}	O^{2-}	F^{-}	Ne
ionic radius				does not form ions easily				does not form ions

Figure 3

(a) Which statement describes how the atomic radius is expected to change across the period?

(1)

☒ **A** decreases across the period

☒ **B** increases across the period

☒ **C** decreases and then increases across the period

☒ **D** increases and then decreases across the period

(b) The electronic configuration of an oxygen atom is $1s^2\ 2s^2\ 2p^4$.

Complete the electronic configuration of the oxide ion, O^{2-}.

(1)

$1s^2\ 2s^2$ ....

(c) Explain the trend in the size of the ionic radius from Li^+ to B^{3+}.

(3)

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(Total for Question 3 = 5 marks)

4 Figure 4 shows a graph of ionisation energy for each successive electron removed from an atom of silicon.

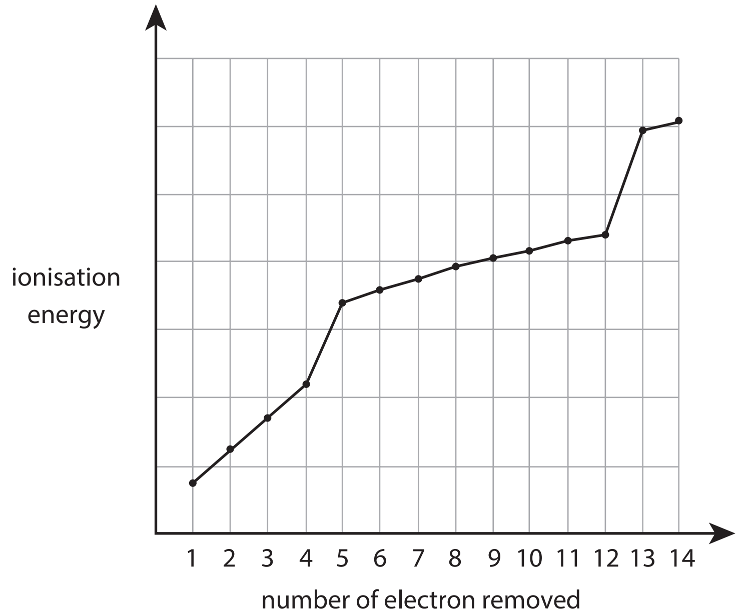

Figure 4

Explain how the pattern of ionisation energies shown in Figure 4 can be used to determine the group and period of the element.

(6)

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(Total for Question 4 = 6 marks)

TOTAL FOR SECTION B = 30 MARKS
TOTAL FOR EXAMINATION = 90 MARKS

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P 7 4 4 9 9 A 0 1 1 1 2

The Periodic Table of Elements

Key

relative atomic mass
atomic symbol
name
atomic (proton) number

1 (1)	2 (2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	3 (13)	4 (14)	5 (15)	6 (16)	7 (17)	0 (8) (18)
							1.0 **H** hydrogen 1										4.0 **He** helium 2
6.9 **Li** lithium 3	9.0 **Be** beryllium 4											10.8 **B** boron 5	12.0 **C** carbon 6	14.0 **N** nitrogen 7	16.0 **O** oxygen 8	19.0 **F** fluorine 9	20.2 **Ne** neon 10
23.0 **Na** sodium 11	24.3 **Mg** magnesium 12											27.0 **Al** aluminium 13	28.1 **Si** silicon 14	31.0 **P** phosphorus 15	32.1 **S** sulfur 16	35.5 **Cl** chlorine 17	39.9 **Ar** argon 18
39.1 **K** potassium 19	40.1 **Ca** calcium 20	45.0 **Sc** scandium 21	47.9 **Ti** titanium 22	50.9 **V** vanadium 23	52.0 **Cr** chromium 24	54.9 **Mn** manganese 25	55.8 **Fe** iron 26	58.9 **Co** cobalt 27	58.7 **Ni** nickel 28	63.5 **Cu** copper 29	65.4 **Zn** zinc 30	69.7 **Ga** gallium 31	72.6 **Ge** germanium 32	74.9 **As** arsenic 33	79.0 **Se** selenium 34	79.9 **Br** bromine 35	83.8 **Kr** krypton 36
85.5 **Rb** rubidium 37	87.6 **Sr** strontium 38	88.9 **Y** yttrium 39	91.2 **Zr** zirconium 40	92.9 **Nb** niobium 41	95.9 **Mo** molybdenum 42	[98] **Tc** technetium 43	101.1 **Ru** ruthenium 44	102.9 **Rh** rhodium 45	106.4 **Pd** palladium 46	107.9 **Ag** silver 47	112.4 **Cd** cadmium 48	114.8 **In** indium 49	118.7 **Sn** tin 50	121.8 **Sb** antimony 51	127.6 **Te** tellurium 52	126.9 **I** iodine 53	131.3 **Xe** xenon 54
132.9 **Cs** caesium 55	137.3 **Ba** barium 56	138.9 **La*** lanthanum 57	178.5 **Hf** hafnium 72	180.9 **Ta** tantalum 73	183.8 **W** tungsten 74	186.2 **Re** rhenium 75	190.2 **Os** osmium 76	192.2 **Ir** iridium 77	195.1 **Pt** platinum 78	197.0 **Au** gold 79	200.6 **Hg** mercury 80	204.4 **Tl** thallium 81	207.2 **Pb** lead 82	209.0 **Bi** bismuth 83	[209] **Po** polonium 84	[210] **At** astatine 85	[222] **Rn** radon 86
[223] **Fr** francium 87	[226] **Ra** radium 88	[227] **Ac*** actinium 89	[261] **Rf** rutherfordium 104	[262] **Db** dubnium 105	[266] **Sg** seaborgium 106	[264] **Bh** bohrium 107	[277] **Hs** hassium 108	[268] **Mt** meitnerium 109	[271] **Ds** darmstadtium 110	[272] **Rg** roentgenium 111							

* Lanthanide series	140 **Ce** cerium 58	141 **Pr** praseodymium 59	144 **Nd** neodymium 60	[147] **Pm** promethium 61	150 **Sm** samarium 62	152 **Eu** europium 63	157 **Gd** gadolinium 64	159 **Tb** terbium 65	163 **Dy** dysprosium 66	165 **Ho** holmium 67	167 **Er** erbium 68	169 **Tm** thulium 69	173 **Yb** ytterbium 70	175 **Lu** lutetium 71
* Actinide series	232 **Th** thorium 90	[231] **Pa** protactinium 91	238 **U** uranium 92	[237] **Np** neptunium 93	[242] **Pu** plutonium 94	[243] **Am** americium 95	[247] **Cm** curium 96	[245] **Bk** berkelium 97	[251] **Cf** californium 98	[254] **Es** einsteinium 99	[253] **Fm** fermium 100	[256] **Md** mendelevium 101	[254] **No** nobelium 102	[257] **Lr** lawrencium 103